

Tasks

1.

In Cisco Packet Tracer, set up a simple network with a router, one internal PC, and one external PC. Assign private IP addresses to the internal PC and public IP addresses to the external PC, then configure static NAT on the router to allow the internal PC to communicate with the external PC.

Ans:

Step 1: Open Cisco Packet Tracer and place one Router, one internal PC (PC-Internal), and one external PC (PC-External) on the workspace.

Step 2: Connect PC-Internal to the router's local interface (GigabitEthernet0/0) and connect PC-External to the router's internet interface (GigabitEthernet0/1).

Step 3: Configure PC-Internal with private IP 192.168.1.10 (Gateway: 192.168.1.1) and PC-External with public IP 203.0.113.10 (Gateway: 203.0.113.1).

Step 4: Click the Router, open the CLI tab, and assign IPs to the interfaces by typing:

```
Router(config)# interface GigabitEthernet0/0
```

```
Router(config-if)# ip address 192.168.1.1 255.255.255.0
```

```
Router(config-if)# ip nat inside
```

```
Router(config-if)# no shutdown
```

```
Router(config-if)# exit
```

Step 5: Configure the outside interface on the same router by typing:

```
Router(config)# interface GigabitEthernet0/1
```

```
Router(config-if)# ip address 203.0.113.1 255.255.255.0
```

```
Router(config-if)# ip nat outside
```

```
Router(config-if)# no shutdown
```

```
Router(config-if)# exit
```

Step 6: Configure the Static NAT mapping to link the private IP to a public IP (203.0.113.5) by typing:

```
Router(config)# ip nat inside source static 192.168.1.10 203.0.113.5
```

2.

Configure Port Address Translation (PAT) on a router in Packet Tracer so that two internal PCs with private IPs can access a simulated public web server using a single public IP address.

Hint: Use the 'ip nat inside source list' command with 'overload' to enable PAT.

Ans:

Step 1: Add a second internal PC (PC-Internal2) to your workspace and connect it to the same local switch or router interface network.

Step 2: Configure PC-Internal2 with the private IP address 192.168.1.20 and set its gateway to 192.168.1.1.

Step 3: Click on the Router, open the CLI, and create an Access Control List (ACL) to match your local network traffic by typing:

```
Router(config)# access-list 1 permit 192.168.1.0 0.0.0.255
```

Step 4: Enable PAT (NAT Overload) using the public router interface (GigabitEthernet0/1) by entering the following command:

```
Router(config)# ip nat inside source list 1 interface GigabitEthernet0/1 overload
```

Step 5: Verify that both local PCs can now simultaneously open connections or ping the simulated public server using just the single public IP address of the router.

3.

Test your NAT configuration by sending a ping from the internal PC to the external PC and capture the NAT translation table on the router to verify the address mapping.

Ans:

Step 1: Click on PC-Internal (192.168.1.10), open its Desktop tab, and open the **Command Prompt**.

Step 2: Type ping 203.0.113.10 and press enter to send test traffic to the external PC.

Step 3: Click on the Router, navigate to the CLI tab, and exit back to privileged EXEC mode by typing exit or end.

Step 4: Run the tracking command by typing:

```
Router# show ip nat translations
```

Step 5: Observe the output table showing the precise mapping between your private IP address (Inside Local) and the translated public IP address (Inside Global).

4.

Modify your Packet Tracer topology to add a third internal PC. Update your NAT or PAT configuration so all three internal PCs can access the external network, and verify that each translation appears in the router's NAT table.

Ans:

Step 1: Drag and drop a third PC (PC-Internal3) into the local office network workspace in Packet Tracer.

Step 2: Assign the private IP address 192.168.1.30 to PC-Internal3 and set its default gateway to 192.168.1.1.

Step 3: Because the ACL configured in Task 2 (192.168.1.0 0.0.0.255) already covers the entire subnet, the third PC is automatically included in the PAT configuration.

Step 4: Open the Command Prompt on PC-Internal3 and generate internet traffic by typing ping 203.0.113.10.

Step 5: Go back to the Router CLI and type show ip nat translations again to verify that all three internal IP addresses now appear active with their unique assigned ports in the table.

5.

Explain in your own words how NAT and PAT help apps like Zomato or Flipkart serve millions of users with limited public IP addresses. Give two real-world scenarios where incorrect NAT/PAT setup could break connectivity for users.

Ans:

How NAT/PAT helps apps like Zomato or Flipkart: There is a global shortage of public IPv4 addresses. Instead of buying a unique public IP for every single employee computer, delivery driver system, or internal server, Zomato and Flipkart use PAT to let thousands of internal office systems safely share a single public IP address to access the internet, saving massive amounts of money and preserving IP addresses.

Real-World Failure Scenario 1 (Wrong Interface Directions): If a network engineer accidentally swaps the configuration commands and types ip nat outside on the internal office port and ip nat inside on the internet port, the router will fail to recognize local traffic, and no user will be able to load the Zomato app or make an order.

Real-World Failure Scenario 2 (NAT Pool Exhaustion / Missing Overload Keyword): If an engineer forgets to type the word overload at the end of the PAT configuration command, the router will perform a strict 1-to-1 conversion. Only the very first user who opens Flipkart will get access; everyone else in the office will get a "Network Connection Timed Out" error because no ports are being shared.